

Epilepsy Detection using Machine Learning on BEED Dataset

Machine Learning Classification Project

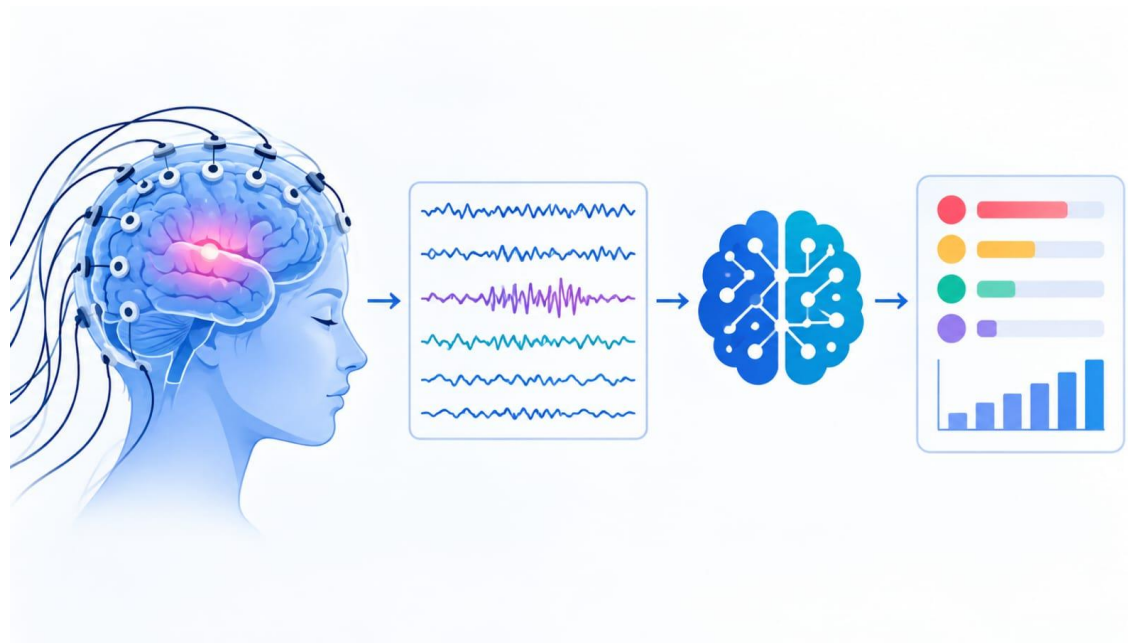
Presented by

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Introduction

- Machine Learning is used to identify patterns in data and make predictions.
- This project uses the Bangalore EEG Epilepsy Dataset (BEED).
- The project applies classification algorithms to EEG-related data.
- The target variable contains four classes: 0, 1, 2 and 3.



Dataset Overview

8,000

Records

16

Input Features

y

Target Variable

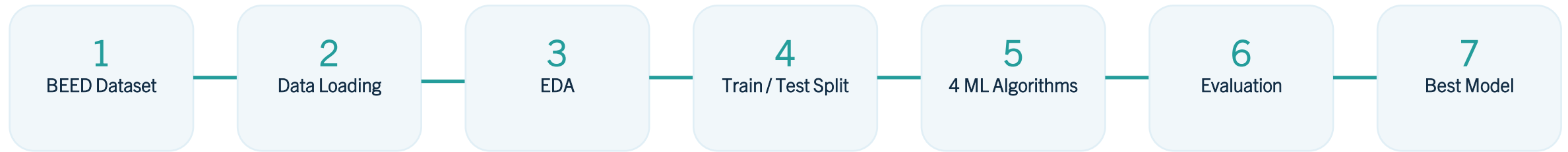
4

Target Classes

- Dataset: Bangalore EEG Epilepsy Dataset (BEED)
- Problem type: Four-class classification
- Feature columns: X1 to X16
- Target column: y



Project Workflow



DATASET	: BEED – Bangalore EEG Epilepsy Dataset
DATA SIZE	: 8,000 records with 16 features
TARGET	: y with 4 classes
DATA ANALYSIS	: Missing values, duplicates, distributions & correlations were analyzed
ML	: 4 classification algorithms were trained and compared
EVALUATION	: Accuracy, Precision, Recall, F1-Score & Confusion Matrix
FINAL GOAL	: Identify the best-performing model for epilepsy classification

Exploratory Data Analysis (EDA)

Data Understanding

Checked shape, information and statistical summary.

Data Quality

Checked missing values and duplicate records.

Target Analysis

Examined class counts and target distribution.

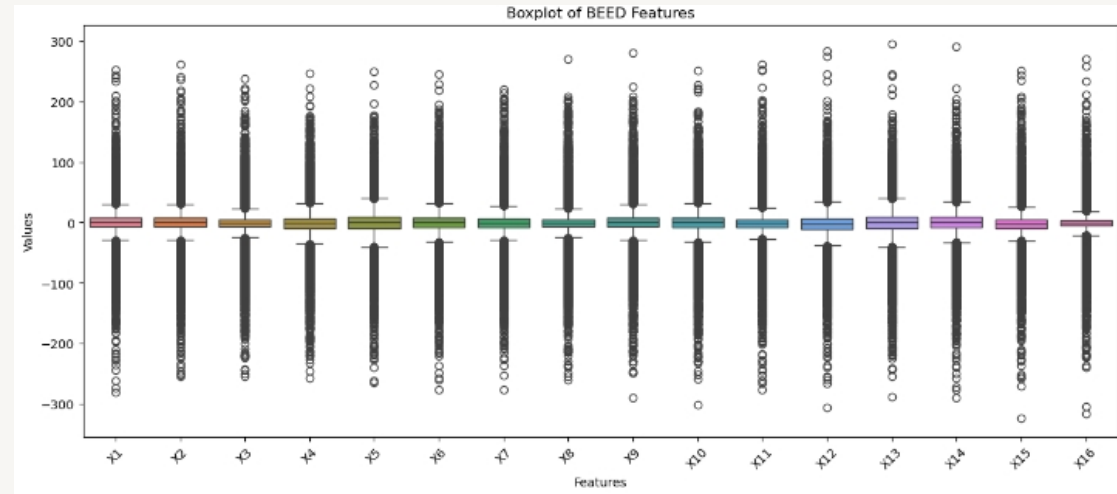
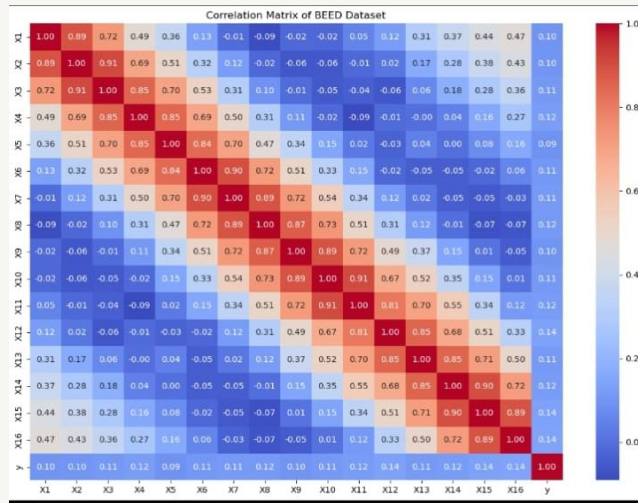
Feature Analysis

Used histograms and boxplots to study feature distributions.

Correlation Analysis

Created a correlation matrix and analyzed feature-to-target relationships.

Correlation & Feature Analysis



- A correlation matrix was created for the numerical features.
- The heatmap helps visualize relationships between variables.
- Feature-to-target correlation was also calculated.
- These analyses help understand patterns in the dataset before modelling.

Machine Learning Algorithms

Logistic Regression

Linear classification model

Decision Tree

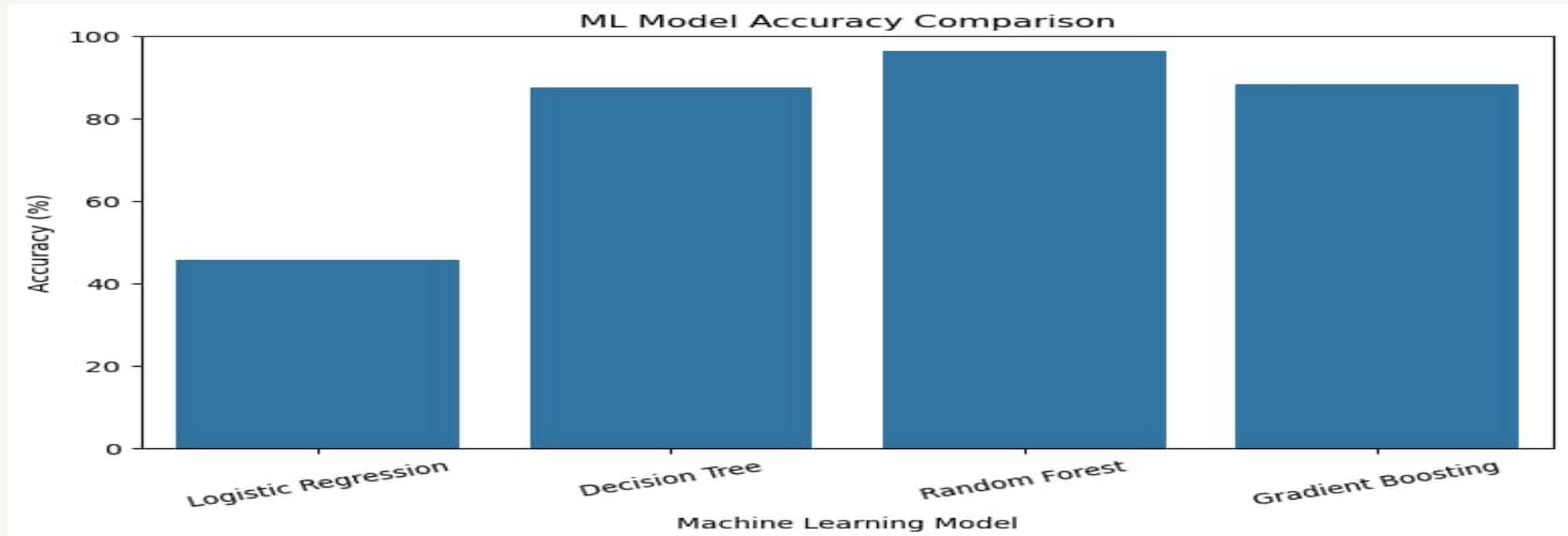
Tree-based classification

Random Forest

Ensemble of decision trees

Gradient Boosting

Sequential boosting ensemble



Model Evaluation

Accuracy

Overall proportion of correct predictions.

Precision

Measures correctness of positive predictions.

Recall

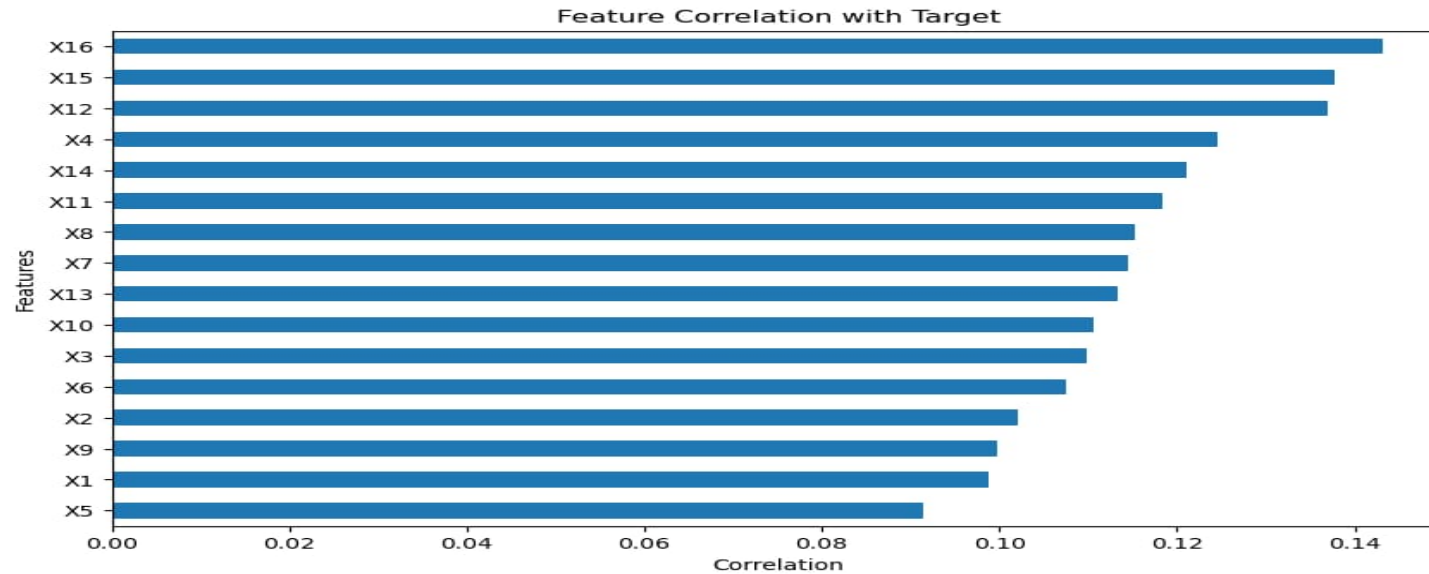
Measures how well actual classes are identified.

F1 Score

Balances precision and recall.

- Classification reports were generated for all four models.
- Confusion matrices were used to compare actual and predicted classes.
- The final model comparison is based on the test dataset.

Results & Feature Importance



- All four models were trained and evaluated using the same train-test split.
- The model with the highest test accuracy is identified automatically in the notebook.
- Random Forest feature importance was analyzed to identify influential features.
- Confusion matrices provide a class-by-class view of model performance.

Conclusion

- The BEED dataset was successfully analyzed using EDA and Machine Learning.
- The project was treated as a four-class classification problem.
- Four classification algorithms were implemented and evaluated.
- The best-performing model was identified using test-set performance.
- Feature importance analysis was used to understand influential features.

Thank You • Jeeva S